

The logo of Gazi University is a circular seal. It features the university's name in Turkish, "GAZİ ÜNİVERSİTESİ", around the top inner edge and the year "1926" at the bottom. In the center is a stylized signature of "Gazi".

GAZİ UNIVERSITY

FACULTY OF ENGINEERING

DEPARTMENT OF INDUSTRIAL ENGINEERING

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IE104-COMPUTER PROGRAMMING I

WEEK 9: LOOPS

Loops

- Loops are code blocks that allow certain tasks to be performed repeatedly within a program. Loops can be both infinite or controlled with a certain condition.
- There are four types of loop structures in the C# language:
 - **for**
 - **foreach**
 - **while**
 - **do while**

For Loop

```
for (expression; expression 2; expression3)
{
    statements;
}
```

- There are three expressions in a for loop separated by two semi-colons(;) .
- Some of these may be empty, but ";" sign must be used.
- **The first statement** is executed only once. It is generally used to define or initialize a loop variable.
- **The second statement** is the part where the loop condition is checked.
- The loop continues as long as the expression produces a “true” value.
- **The last statement** usually is the part where the value of the loop variable is changed.

For Loop

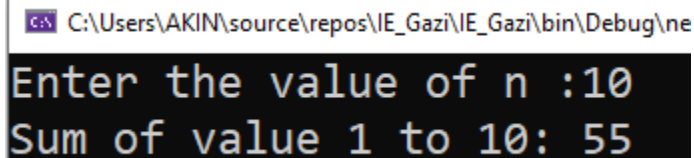
```
for(expr1;expr2;expr3)
{
    statements;
}
```

Usage:

- `for(;;)` is an infinite loop.
- `for(int i=1;i<=100;i++)`
- `for(int i=1,j=10;(i<=10)&(j>=1);i++,j--)`

For Loop-Example

- Write a C# program that finds the sum of all numbers from 1 up to n which is entered from the keyboard.
- Expected screen is shown below:



```
C:\Users\AKIN\source\repos\IE_Gazi\IE_Gazi\bin\Debug\ne
Enter the value of n :10
Sum of value 1 to 10: 55
```

```
static void Main(string[] args)
{
    // Example for Gazi IE
    int i, n, sum = 0;
    Console.Write("Enter the value of n :");

    n = Convert.ToInt32(Console.ReadLine());

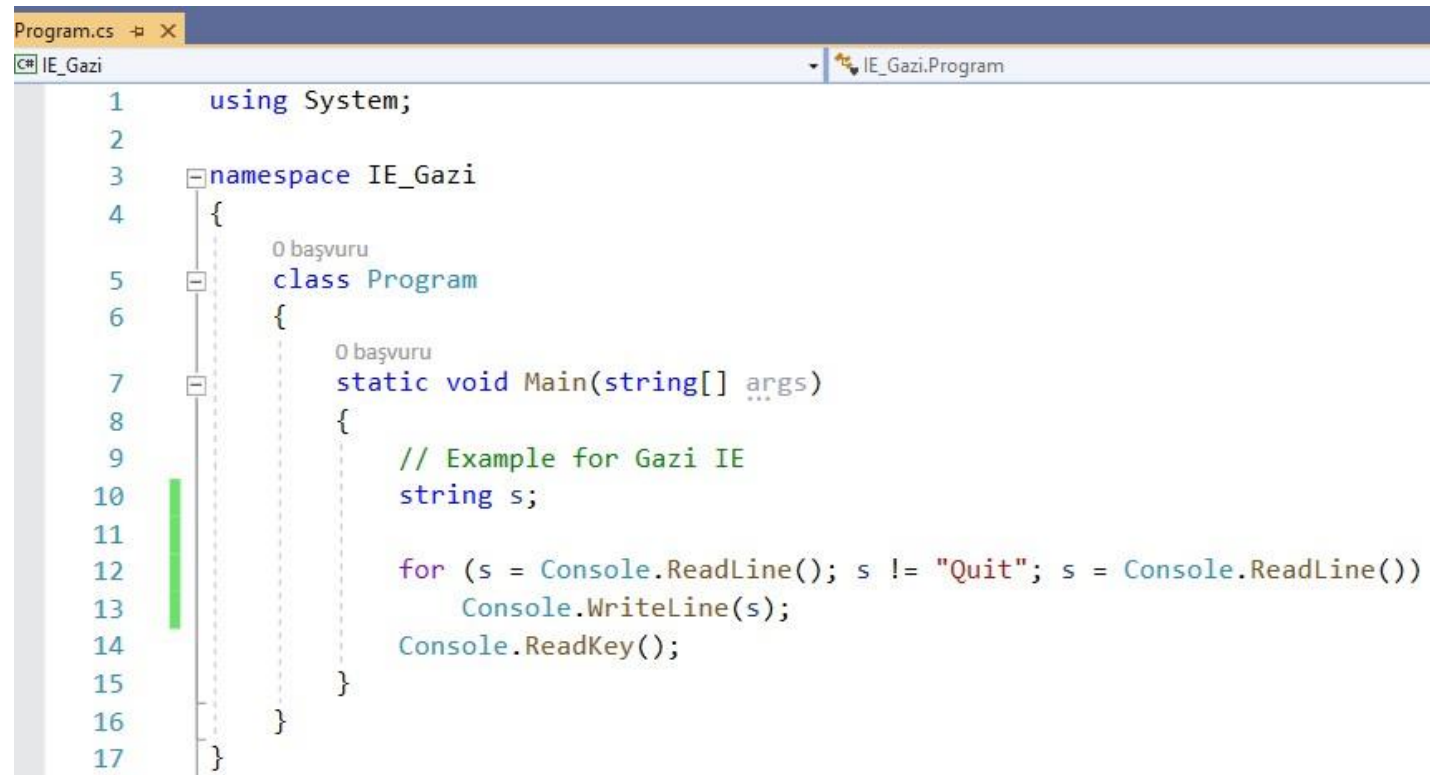
    for (i = 1; i <= n; i++)
    {
        sum += i;
    }

    Console.WriteLine("Sum of value 1 to {0}: {1} ", n, sum);

    Console.ReadKey();
}
```

For Loop-Example

- Write a C# program that quits when user enters the word "Quit"



```
Program.cs [X]
C# IE_Gazi IE_Gazi.Program
1 using System;
2
3 namespace IE_Gazi
4 {
5     class Program
6     {
7         static void Main(string[] args)
8         {
9             // Example for Gazi IE
10            string s;
11
12            for (s = Console.ReadLine(); s != "Quit"; s = Console.ReadLine())
13                Console.WriteLine(s);
14            Console.ReadKey();
15        }
16    }
17 }
```

For Loop

- Write a C# program that displays the numbers from 0 to n according to the incremental value entered from keyboard.
- Expected screen is shown below:

```
C:\Users\AKIN\source\repos\IE_Gazi\IE_Gazi\bin\Debug\netcoreapp3.1'
Enter the value of n :7
Enter the incremental value:3
0
3
6
```

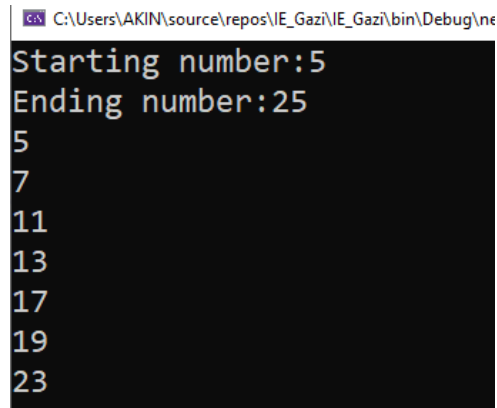
```
static void Main(string[] args)
{
    // Example for Gazi IE
    int i=0, n, inc;
    Console.Write("Enter the value of n :");
    n = Convert.ToInt32(Console.ReadLine());

    Console.Write("Enter the incremental value:");
    inc = Convert.ToInt32(Console.ReadLine());

    for (; i < n; )
    {
        Console.WriteLine("{0} ", i);
        i += inc;
    }
    Console.ReadKey();
}
```

For Loop

- Write a C# program that finds the prime numbers from a starting number to the ending number and prints it to the screen.
- Expected screen is shown below:



A screenshot of a console application window. The title bar shows the file path: C:\Users\AKIN\source\repos\IE_Gazi\IE_Gazi\bin\Debug\ne. The console output is as follows:

```
Starting number:5
Ending number:25
5
7
11
13
17
19
23
```

```
static void Main(string[] args)
{
    // Example for Gazi IE

    int k, t, sum, n1, n2;

    Console.Write("Starting number:");
    n1 = Convert.ToInt32(Console.ReadLine());
    Console.Write("Ending number:");
    n2 = Convert.ToInt32(Console.ReadLine());
    if (n1 <= n2)
    {
        for (k = n1; k <= n2; k++)
        {
            sum = 0;
            for (t = 1; t <= k; t++)
            {
                if (k % t == 0)
                    sum += t;
            }
            if (sum == k + 1)
                Console.WriteLine(k);
        }
    }
    else
        Console.WriteLine("Please enter a valid interval...");
    Console.ReadKey();
}
```


For Loop

- Alternative solution:
- Expected screen is shown below:

```
C:\Users\AKIN\source\repos\IE_Gazi\IE_Gazi\bin\Debug\net
Starting number:5
Ending number:25
5
7
11
13
17
19
23
```

```
static void Main(string[] args)
{
    // Example for Gazi IE
    int i, j, n1, n2, flag;

    Console.Write("Starting number:");
    n1 = Convert.ToInt32(Console.ReadLine());
    Console.Write("Ending number:");
    n2 = Convert.ToInt32(Console.ReadLine());
    if (n1 <= n2)
    {
        for (i = n1; i <= n2; i++)
        {
            flag = 0;
            for (j = 2; j < i; j++)
            {
                if (i % j == 0)
                { flag = 1; break; }
            }
            if (flag==0)
            {
                Console.WriteLine(i);
            }
        }
    }
    else
        Console.WriteLine("Please enter a valid interval...");
    Console.ReadKey();
}
```

While loop

```
while (condition)
{
    statements
}
```

- The while loop continues to run as long as the specified condition is true. If you don't set a condition that makes while condition check result false or if you forget to enter a break statement, while loop will run infinitely that might cause your program to **freeze!**.

While and do-while Loop

```
do while  
{  
    statements  
} while(condition);
```

In some cases, the loop may need to be run at least once. In this case «do while» can be used.

IMPORTANT NOTE: If the condition check in **for and **while** loops is not satisfied, the loop statement block is not executed at all.**

While and do-while Loop

C:\Users\AKIN\source\repos\IE_Gazi\IE_Gazi\bin\Debug\netcoreapp3.1\IE_Gazi.exe

```
5 10 15 20 25 30 35 40 45 50
55 60 65 70 75 80 85 90 95 100
105 110 115 120 125 130 135 140 145 150
155 160 165 170 175 180 185 190 195 200
205 210 215 220 225 230 235 240 245 250
255 260 265 270 275 280 285 290 295 300
305 310 315 320 325 330 335 340 345 350
355 360 365 370 375 380 385 390 395 400
405 410 415 420 425 430 435 440 445 450
455 460 465 470 475 480 485 490 495 500
505 510 515 520 525 530 535 540 545 550
555 560 565 570 575 580 585 590 595 600
605 610 615 620 625 630 635 640 645 650
655 660 665 670 675 680 685 690 695 700
705 710 715 720 725 730 735 740 745 750
755 760 765 770 775 780 785 790 795 800
805 810 815 820 825 830 835 840 845 850
855 860 865 870 875 880 885 890 895 900
905 910 915 920 925 930 935 940 945 950
955 960 965 970 975 980 985 990 995 1000
```

Write a program that outputs the values above. Make sure to cut the print statement when the number reaches 50 or its multipliers.

```
static void Main(string[] args)
{
    // Example for Gazi IE
    int i = 0;
    while (i < 1000)
    {
        i += 5;
        Console.Write("{0,5}", i);
        if (i % 50 == 0)
            Console.WriteLine();
    }

    Console.ReadKey();
}
```

While and do-while Loop

Write a program that does simple calculator operations. Make sure to prompt user a menu of options as shown in the image and do the operation when user selects an option. You can use switch statement or if else statement.

```
C:\Users\AKIN\source\repos\IE_Gazi\IE_Gazi\bin\Debug\netcoreapp3.1\IE_
Operations
=====
1 - Addition
2 - Subtraction
3 - Multiplication
4 - Division
0 - Quit

Choose your operation:9
Wrong Selection.. Try again please!

Operations
=====
1 - Addition
2 - Subtraction
3 - Multiplication
4 - Division
0 - Quit

Choose your operation:1
Addition is selected.

Operations
=====
1 - Addition
2 - Subtraction
3 - Multiplication
4 - Division
0 - Quit

Choose your operation:0
You just wanted to quit.
```

```
static void Main(string[] args)
{
    // Example for Gazi IE
    int c;
    do
    {
        Console.WriteLine("Operations\n=====");
        Console.WriteLine("1 - Addition");
        Console.WriteLine("2 - Subtraction");
        Console.WriteLine("3 - Multiplication");
        Console.WriteLine("4 - Division");
        Console.WriteLine("0 - Quit\n");
        Console.Write("Choose your operation:");
        c = Convert.ToInt32(Console.ReadLine());
        switch (c)
        {
            case 1: Console.WriteLine("Addition is selected.\n"); break;
            case 2: Console.WriteLine("Subtraction is selected.\n"); break;
            case 3: Console.WriteLine("Multiplication is selected.\n"); break;
            case 4: Console.WriteLine("Division is selected.\n"); break;
            case 0: Console.WriteLine("You just wanted to quit.\n"); break;
            default:
                Console.WriteLine("Wrong Selection.. Try again please!\n"); break;
        }
    } while (c != 0);

    Console.ReadKey();
}
```

While and do-while Loop

Alternative solution

```
1 using System;
2
3 class Program
4 {
5     static void Main()
6     {
7         int choice;
8         do
9         {
10             Console.WriteLine("===== SIMPLE CALCULATOR =====");
11             Console.WriteLine("1. Addition");
12             Console.WriteLine("2. Subtraction");
13             Console.WriteLine("3. Multiplication");
14             Console.WriteLine("4. Division");
15             Console.WriteLine("5. Exit");
16             Console.Write("Enter your choice: ");
17             choice = Convert.ToInt32(Console.ReadLine());
18             if (choice >= 1 && choice <= 4)
19             {
20                 Console.Write("Enter first number: ");
21                 double num1 = Convert.ToDouble(Console.ReadLine());
22                 Console.Write("Enter second number: ");
23                 double num2 = Convert.ToDouble(Console.ReadLine());
```

```
24             switch (choice)
25             {
26                 case 1:
27                     Console.WriteLine("Result = " + (num1 + num2));
28                     break;
29                 case 2:
30                     Console.WriteLine("Result = " + (num1 - num2));
31                     break;
32                 case 3:
33                     Console.WriteLine("Result = " + (num1 * num2));
34                     break;
35                 case 4:
36                     if (num2 != 0)
37                         Console.WriteLine("Result = " + (num1 / num2));
38                     else
39                         Console.WriteLine("Error: Division by zero is not
40                                     allowed.");
41                     break;
42             }
43             else if (choice == 5)
44             {
45                 Console.WriteLine("Exiting program...");
46             }
47             else
48             {
49                 Console.WriteLine("Invalid choice. Please select from 1 to 5.");
50             }
51             Console.WriteLine();
52         } while (choice != 5);
53     }
54 }
```

Foreach Loop

- Different than the other loop structures.
- Provides step-by-step navigation through collection-based objects.
- This loop provides an access to individual elements one by one, but the accessed elements are read-only (not changeable).

Foreach Loop

```
1 using System;
2
3 class Program
4 {
5     static void Main()
6     {
7         // For loop example
8         Console.WriteLine("For loop example:");
9         for (int number = 1; number <= 5; number++)
10        {
11            Console.WriteLine("Number: " + number);
12        }
13
14        Console.WriteLine();
15
16        // Foreach loop example
17        Console.WriteLine("Foreach loop example:");
18        string[] items = { "Pen", "Book", "Bag", "Pencil" };
19
20        foreach (string item in items)
21        {
22            Console.WriteLine("Item: " + item);
23        }
24    }
25 }
```

Output

For loop example:

Number: 1

Number: 2

Number: 3

Number: 4

Number: 5

Foreach loop example:

Item: Pen

Item: Book

Item: Bag

Item: Pencil

Jump Statements

- **Break:** A running loop can be stopped by using the word “**break**.” After that, program execution continues with the lines that come after the loop. The word **break** can only be used in loops and switch statements.
- **Continue:** Allows the loop to move on to the next **iteration**.
- **Goto:** It is used to jump to another part of the program marked by a label. It is not very common in object-oriented programming and should be avoided as much as possible, except in a switch statement.
- **Return:** It is used to end a method at any time. Once the method ends, program flow continues from the function which the method was called.

Jump Statements- Example

```
1  using System;
2
3  class Program
4  {
5      static void Main()
6      {
7          for (int number = 1; number <= 5; number++)
8          {
9              if (number == 2)
10                 continue;
11
12                 if (number == 4)
13                     break;
14
15                 Console.WriteLine("Number: " + number);
16             }
17
18             goto End;
19
20         End:
21             Console.WriteLine("Goto statement used.");
22             ShowMessage();
23         }
24
25         static void ShowMessage()
26         {
27             Console.WriteLine("Return statement used.");
28             return;
29         }
30     }
```

Jump Statements-Exercise

- Write a C# program that sums the negative numbers the user entered.
 - Condition - 1: The program will print "Enter a negative number:" and process number till the user enters 0.
 - Condition - 2: If the user enters a positive number, print a warning message then demand to enter a negative number again.
 - The expected screen is shown below:

C:\Users\AKIN\source\repos\IE_Gazi\IE_Gazi\bin\Debug\netcoreapp3.1\IE_Gazi.exe

```
Enter a negative number:6
You have entered a positive number. Try again!
Enter a negative number:-3
Enter a negative number:-4
Enter a negative number:-1
Enter a negative number:0
You have entered 0. Now the program terminates.
Sum of negative number you entered:-8
```

```
// Example for Gazi IE
0 başvuru
static void Main(string[] args)
{
    int num, sum = 0;

    while (true)
    {
        Console.WriteLine("Enter a negative number:");
        num = int.Parse(Console.ReadLine());

        if (num > 0)
        {
            Console.WriteLine("You have entered a positive number. Try again!");
            continue;
        }
        if (num == 0)
        {
            Console.WriteLine("You have entered 0. Now the program terminates.");
            break;
        }

        sum += num;
    }

    Console.WriteLine("Sum of the negative numbers you entered:{0}", sum);
}
```

THE END



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GOT ANY QUESTIONS LATER?

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